

PARAMEDIC PROGRAM · PHARMACOLOGY

Paramedic Pharmacology Workbook

Drug math, IV fluids and infusion pumps, explained step by step with formulas, worked examples and practice problems.

NAME _____

COHORT _____

INSTRUCTOR _____

How to use this workbook

Each chapter starts with the concept, then gives the formula in a labeled box, then walks through worked examples one step at a time. Practice problems follow, with space to show your work. Worked solutions are in the answer key at the back.

Show every step. On a call, a calculation you can check is worth more than a fast guess, and your partner can only double-check work they can see.

> **LOCAL PROTOCOL RULES**
>
> Drugs, concentrations and doses here are teaching examples drawn from common EMS practice. Your formulary, packaging and medical director's protocols always win. Check every vial label against the order before you draw up.

Contents

01 Medication safety foundations

02 The metric system and conversions

03 IV fluids in the paramedic scope

04 Dose and volume: how much to draw up

05 Weight-based calculations

06 Drip rates I: volume over time

07 Drip rates II: concentration infusions

08 Percent and ratio concentrations

09 IV infusion pumps

10 Formula reference card

11 Answer key with worked solutions

CHAPTER ONE

01 Medication safety foundations

Drug math only matters if it lands in the right patient, by the right route, at the right time. Most medication errors in EMS are not arithmetic failures. They come from grabbing the wrong vial, misreading a concentration, or a decimal point in the wrong place. Build the habits in this chapter before you build speed.

THE RIGHTS OF MEDICATION ADMINISTRATION

1. **Right patient**: confirm identity and allergies.
2. **Right medication**: read the label three times (when you pick it up, draw it up, and give it).
3. **Right dose**: calculate, then have a partner verify.
4. **Right route**: IV, IO, IM, IN, SL, PO, neb, ET.
5. **Right time**: rate of push, interval between doses.
6. **Right documentation**: drug, dose, route, time, response.
7. **Right reason, response, and right to refuse**.

WRITING NUMBERS SAFELY

WRITE	NEVER	WHY
`0.5 mg`	`.5 mg`	Missed decimal reads as 5 mg (10× overdose)
`5 mg`	`5.0 mg`	Trailing zero reads as 50 mg
`mcg`	`µg`	µ looks like "m": 1,000× error
`units`	`U, IU`	U reads as 0 or 4
`mL`	`cc`	cc reads as "u" or "00"

Rounding rules used in this workbook

QUANTITY	ROUND TO	EXAMPLE
Volume to draw up, over 1 mL	Tenths (0.1 mL)	`2.36 mL → 2.4 mL`
Volume to draw up, under 1 mL (1 mL syringe)	Hundredths (0.01 mL)	`0.456 mL → 0.46 mL`
Patient weight in kg	Tenths, or whole kg for adults	`16.36 kg → 16.4 kg`
Drip rate (gtt/min)	Whole drops	`31.25 gtt/min → 31 gtt/min`
Pump rate (mL/hr)	Tenths or whole, per pump	`26.25 mL/hr → 26.3 mL/hr`

Round only at the end. Carry full numbers through each step so small rounding errors do not stack up.

HIGH-ALERT MEDICATIONS

Vasopressors (norepinephrine, epinephrine infusions, dopamine), opioids, sedatives, insulin, potassium and concentrated electrolytes cause serious harm in small errors. Use an independent double-check: your partner calculates the dose separately before you compare answers. Reading your answer aloud for them to agree with does not count as a check.

FIELD TIP: DOES THE ANSWER MAKE SENSE?

Before you push, ask whether the number is plausible. Most single IV push doses fit in 0.1 to 10 mL. If your answer is 25 mL of fentanyl or 0.02 mL of naloxone, stop and recalculate.

CHAPTER TWO

02 The metric system and conversions

Almost every drug calculation starts with getting the order and the vial into the **same unit**. If the order is in micrograms and the vial is labeled in milligrams, convert first. The metric system makes this easy: each common step is a factor of 1,000.

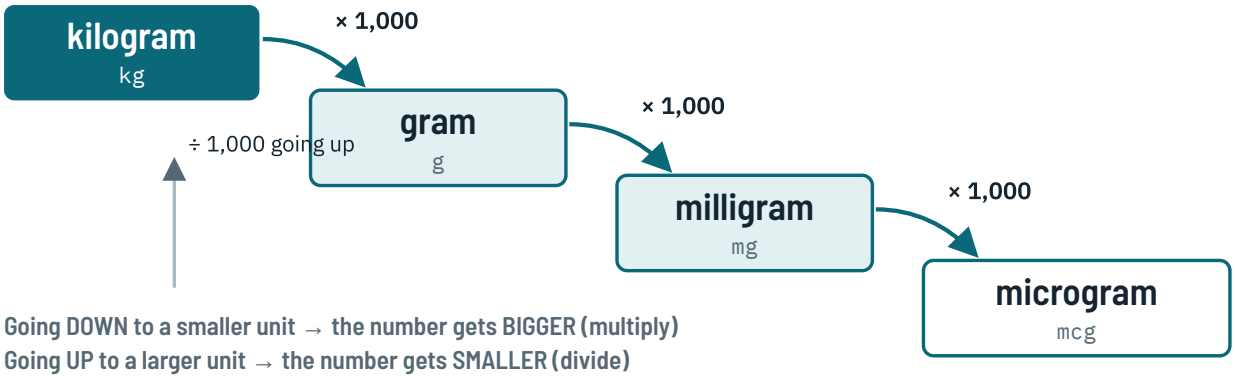


FIGURE 2.1 The metric staircase. Volume works the same way: 1 L = 1,000 mL.

CONVERSION FACTORS TO MEMORIZE — **FORMULA 2.1**

Mass	`1 kg = 1,000 g`	`1 g = 1,000 mg`	`1 mg = 1,000 mcg`
Volume	`1 L = 1,000 mL`	`1 mL = 1 cc`	`1 tsp ≈ 5 mL`
Weight	`1 kg = 2.2 lb`	`lb ÷ 2.2 = kg`	`kg × 2.2 = lb`
Time	`1 hr = 60 min`	`mL/hr ÷ 60 = mL/min`	`mL/min × 60 = mL/`
Temperature	`°C = (°F − 32) ÷ 1.8`	`°F = (°C × 1.8) + 32`	

The decimal-shift method

Multiplying or dividing by 1,000 means moving the decimal point three places. Move it **right** when going to a smaller unit, **left** when going to a larger unit.

WORKED EXAMPLE 2.1 0.25 mg to mcg

DIRECTION

mg → mcg is a step down, so multiply by 1,000.

SHIFT

`0.250 → 250.`

ANSWER

250 mcg

WORKED EXAMPLE 2.2 176 lb to kg

RULE

Pounds to kilograms: divide by 2.2.

MATH

`176 ÷ 2.2 = 80`

ANSWER

80 kg

WORKED EXAMPLE 2.3 1.5 g to mg

DIRECTION

g → mg is a step down: multiply by 1,000.

SHIFT

`1.500 → 1500.`

ANSWER

1,500 mg

WORKED EXAMPLE 2.4 102.2 °F to °C

SUBTRACT

`102.2 − 32 = 70.2`

DIVIDE

`70.2 ÷ 1.8 = 39`

ANSWER

39 °C

FIELD TIP: QUICK KG ESTIMATE

Halve the pounds, then subtract 10% of that. 180 lb → 90 → 90 − 9 = 81 kg (exact: 81.8 kg). Use this to sanity-check, then calculate exactly for the dose.

Practice 2: Conversions

Answers: Key 2

2.1 0.4 mg = _____ mcg

2.2 2,500 mcg = _____ mg

2.3 198 lb = _____ kg

2.4 1.2 L = _____ mL

2.5 A newborn weighs 3.5 kg. How many pounds is that? _____ lb

2.6 0.025 g = _____ mg

2.7 A patient's temperature is 39.5 °C. Convert to °F. _____ °F

2.8 An infusion runs at 90 mL/hr. How many mL per minute is that? _____ mL/min

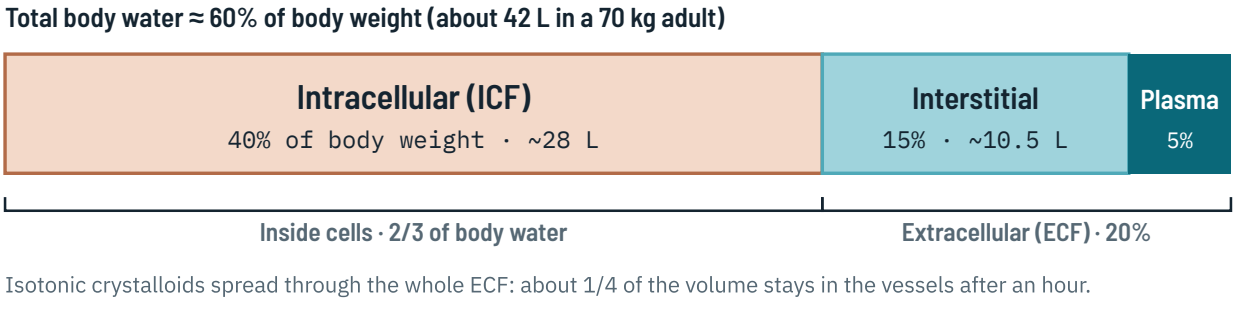
CHAPTER THREE

03 IV fluids in the paramedic scope

IV fluids are drugs. Each one has indications, contraindications and a dose. To choose well you need two ideas: **where water lives in the body**, and **tonicity**, which decides which way water moves once a fluid is in the bloodstream.

Body fluid compartments

About 60% of an adult's body weight is water. Two-thirds of it sits inside cells (intracellular fluid, ICF). The remaining third is extracellular (ECF), split between the interstitial space around the cells and the plasma inside the vessels.



Isotonic crystalloids spread through the whole ECF: about 1/4 of the volume stays in the vessels after an hour.

FIGURE 3.1 This is why a 1 L saline bolus raises blood volume by only about 250 mL once it equilibrates, and why reassessment after each bolus matters.

Tonicity: which way does water move?

Tonicity compares a fluid's effective solute concentration to plasma (about 275–295 mOsm/kg). Water always moves toward the side with **more** solute.

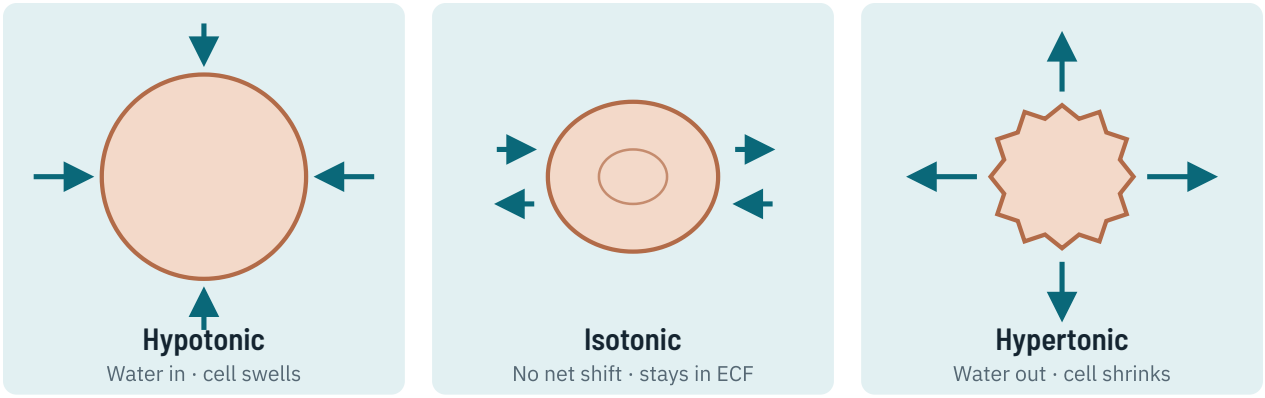


FIGURE 3.2 Isotonic fluids expand the extracellular space and are the workhorse for volume replacement. Hypotonic fluids shift water into cells, including brain cells, which is why they are avoided in head injury.

The fluids you will hang

Most paramedic formularies carry normal saline and lactated Ringer's for volume, D5W as a drug diluent, and D10 for hypoglycemia. Some services also carry a balanced solution or monitor hypotonic maintenance fluids on interfacility transfers. Bags come in 100, 250, 500 and 1,000 mL sizes.

0.9% Sodium Chloride

"Normal saline" · NS

ISOTONIC CRYSTALLOID

PER LITER

Na^+ 154 · Cl^- 154 mEq

OSMOLARITY

308 mOsm/L · pH ≈ 5.5

USES

Hypovolemia, dehydration, sepsis, DKA, heat illness, burns, crush injury. Default carrier for most IV drugs. The only crystalloid compatible with blood products.

CAUTIONS

Fluid overload (CHF, renal failure, pulmonary edema). Large volumes cause hyperchloremic metabolic acidosis.

Lactated Ringer's

LR · Ringer's lactate · Hartmann's

ISOTONIC | BALANCED

PER LITER

Na^+ 130 · K^+ 4 · Ca^{2+} 2.7 · Cl^- 109 · lactate 28 mEq

OSMOLARITY

273 mOsm/L · pH ≈ 6.5

USES

Trauma and hemorrhage until blood is available, burns (Parkland formula), large-volume resuscitation. Electrolytes closer to plasma than NS.

CAUTIONS

Calcium binds the citrate in stored blood: do not run in the same line as blood. Liver failure (lactate is cleared by the liver). Mildly hypotonic: many protocols prefer NS in traumatic brain injury.

5% Dextrose in Water

D5W

ISOTONIC IN THE BAG | HYPOTONIC IN THE BODY

PER LITER

Dextrose 50 g · 170 kcal

OSMOLARITY

252 mOsm/L · pH ≈ 4.3

USES

Diluent for drugs that require it (amiodarone infusions in many protocols). KVO line. Not a volume replacement fluid.

CAUTIONS

The cells use the dextrose, leaving free water that moves into cells. Avoid in head injury, stroke and hyperglycemia. Will not raise blood pressure.

10% Dextrose in Water

D10 · D10W

HYPERTONIC | TREATS HYPOGLYCEMIA

PER LITER

Dextrose 100 g = 0.1 g/mL

OSMOLARITY

505 mOsm/L

USES

Symptomatic hypoglycemia in adults and children, titrated in 5–25 g aliquots (50–250 mL). Less vein damage than D50 if it infiltrates.

CAUTIONS

Recheck glucose after each aliquot. Watch the IV site: extravasation still damages tissue.

0.45% Sodium Chloride

"Half-normal saline" · ½NS

HYPOTONIC | SCOPE VARIES

PER LITER

Na^+ 77 · Cl^- 77 mEq

OSMOLARITY

154 mOsm/L

USES

Maintenance fluid seen on interfacility transfers, sometimes with dextrose (D5 ½NS) or KCl added. Usually monitored, not started, by paramedics.

CAUTIONS

Never for volume resuscitation. Avoid in head injury, burns, trauma and hyponatremia.

Plasma-Lyte A / Normosol-R

Balanced multiple-electrolyte solution

ISOTONIC | CARRIED BY SOME SERVICES

PER LITER

Na^+ 140 · K^+ 5 · Mg^{2+} 3 · Cl^- 98 · acetate 27 · gluconate 23 mEq

OSMOLARITY

294 mOsm/L · pH ≈ 7.4

USES

Volume replacement with less acid-base disturbance than NS. No calcium, so it is compatible with blood products.

CAUTIONS

Fluid overload; contains potassium (caution in hyperkalemia).

COLLOIDS AND BLOOD

Colloids (albumin, hetastarch) contain large molecules that stay in the vessels. Blood products (packed red cells, plasma, whole blood) are the preferred replacement for hemorrhage. Most paramedics only monitor these during transfers; some critical care and prehospital blood programs start them under specific protocols.

Fluid dosing in the field

SITUATION	TYPICAL APPROACH	REASSESS FOR
Adult hypovolemia, dehydration, sepsis	Isotonic crystalloid in 250–500 mL boluses. Sepsis protocols may target up to 30 mL/kg.	BP, HR, mental status, lung sounds, SpO_2
Hemorrhagic shock (no TBI)	Permissive hypotension: small boluses to a radial pulse or SBP about 80–90 mmHg. Blood first where available.	Radial pulse, mentation
Traumatic brain injury	Avoid any hypotension. Many protocols target SBP ≥ 110 mmHg. Isotonic fluid only.	GCS, pupils, BP
Pediatric hypovolemia	20 mL/kg isotonic bolus, repeat after reassessment. 10 mL/kg for neonates or suspected cardiac dysfunction.	Cap refill, HR, mental status, liver edge, crackles
Burns	LR by formula: 2–4 mL × kg × %TBSA over 24 h, half in the first 8 h from the time of the burn.	Urine output on long transports
KVO / saline lock	Run just enough to keep the catheter open, often 10–30 mL/hr.	Site patency

WORKED EXAMPLE 3.1

Parkland formula: 80 kg adult, 30% TBSA partial and full thickness burns

24-HOUR

4 mL × 80 kg × 30 = 9,600 mL LR

FIRST 8 H

9,600 ÷ 2 = 4,800 mL

RATE

4,800 mL ÷ 8 hr = 600 mL/hr

ANSWER

600 mL/hr (if the clock starts at the time of injury; subtract fluid already given). Many burn centers now start at 2 mL/kg/%TBSA: 300 mL/hr.

Catheters and administration sets

Flow through a catheter depends mostly on its **diameter** and **length** (Poiseuille's law). Doubling the radius raises flow sixteen-fold, so a short, large-bore catheter is what you want for volume. Gauge numbers run backwards: the smaller the number, the bigger the catheter.

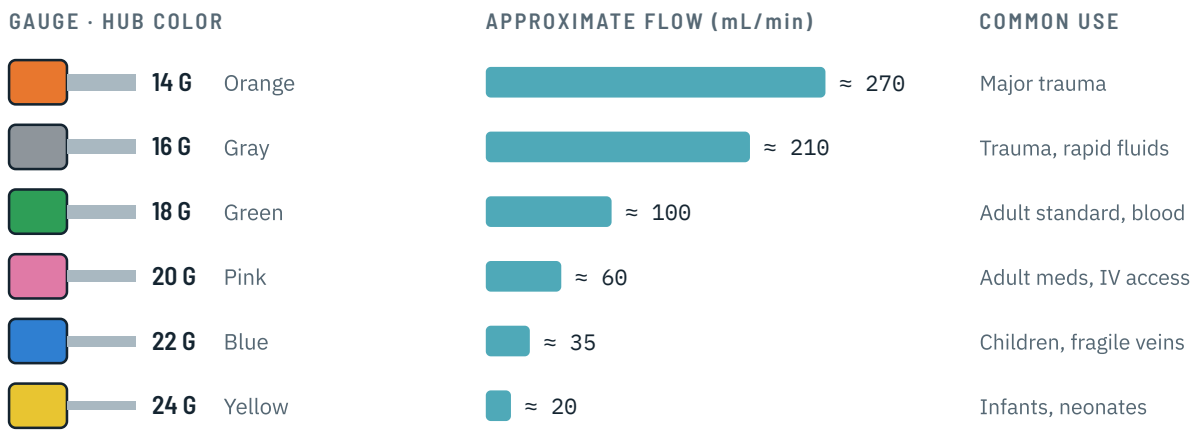
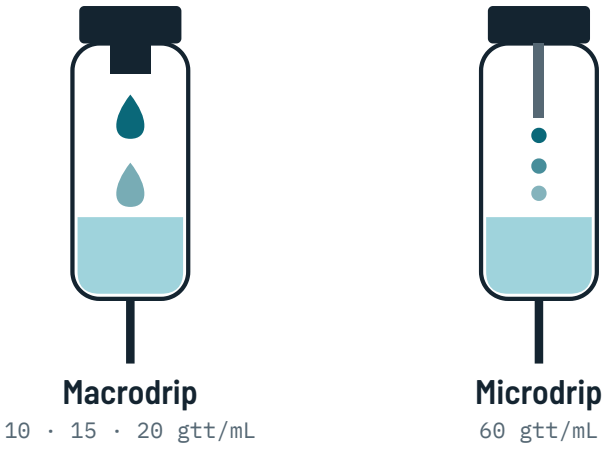


FIGURE 3.3 ISO 10555 hub colors. Flow rates are approximate for a 1–1.25 inch catheter under gravity with crystalloid; they vary by length and manufacturer.



Drop factor (gtt/mL)

Printed on the tubing package. Always read it; never assume.

Use macrodrip for

Fluid boluses, trauma, burns, blood (10 gtt/mL filtered set).

Use microdrip for

Medication infusions, children, precise low rates, KVO.

FIGURE 3.4 Inside a microdrip chamber you can see a thin metal cannula. With a 60 gtt/mL set, gtt/min equals mL/hr, which makes gravity medication drips simple to set.

Practice 3: IV fluids

Answers: Key 3

3.1 Which crystalloid is the only one compatible with blood products, and why can't LR run in the same line?

3.2 D5W is isotonic in the bag. Explain why it behaves as a hypotonic fluid once infused.

3.3 Name the tonicity of each: 0.9% NaCl _____ · 0.45% NaCl _____ · D10W _____

3.4 A 25 kg child is in compensated hypovolemic shock. Calculate a 20 mL/kg bolus. _____ mL

3.5 Your patient has a severe head injury and is hypotensive. Which fluids should you avoid, and why?

3.6 A 60 kg adult has 40% TBSA burns. Using 4 mL/kg/%TBSA, what rate (mL/hr) runs for the first 8 hours, starting at the time of injury? _____ mL/hr

3.7 About how much of a 1,000 mL NS bolus remains in the vascular space after an hour? _____ mL

CHAPTER FOUR

04 Dose and volume: how much to draw up

The order tells you the **dose** (mg, mcg, g, units). The vial tells you the **concentration**: how much drug is in each mL. Your job is to find the **volume** that holds the ordered dose.

DESIRED OVER HAVE — **FORMULA 4.1**

$$\text{Volume to give (mL)} = \frac{D}{H} \times Q$$

D desired dose (the order) **H** dose on hand (drug amount on the label)

Q quantity (the volume that holds H)

D and H must be in the same unit. Convert first if they are not.

WORKED EXAMPLE 4.1 Fentanyl 75 mcg IV. Vial: 100 mcg / 2 mL

IDENTIFY

D = 75 mcg · H = 100 mcg · Q = 2 mL

UNITS

Both in mcg. No conversion needed.

SOLVE

75 ÷ 100 × 2 = 0.75 × 2 = 1.5 mL

CHECK

75 is a bit less than 100, so the answer should be a bit less than 2 mL. It is.

ANSWER

Draw up 1.5 mL

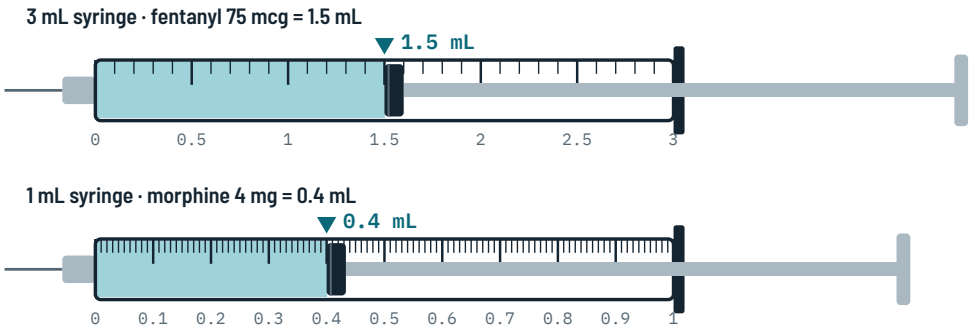


FIGURE 4.1 Read the volume at the leading edge of the rubber stopper, the ring closest to the needle. Pick the smallest syringe that holds the volume: it has the finest markings.

Three ways to solve the same problem

Use whichever method you trust. They always agree. Here is the same order solved three ways: **morphine 4 mg IV, vial 10 mg/mL**.

METHOD A Formula

`D/H × Q`

`4 ÷ 10 × 1 mL`

`0.4 mL`

METHOD B Ratio and proportion

`10 mg : 1 mL = 4 mg : x mL`

`10x = 4 → x = 0.4`

`0.4 mL`

METHOD C

Dimensional analysis

`4 mg × (1 mL / 10 mg)`

`mg cancels → 0.4 mL`

`0.4 mL`

When the units don't match

WORKED EXAMPLE 4.2

Order: naloxone 400 mcg IV. Vial: 2 mg / 2 mL

CONVERT

`400 mcg = 0.4 mg`

IDENTIFY

`D = 0.4 mg · H = 2 mg · Q = 2 mL`

SOLVE

`0.4 ÷ 2 × 2 = 0.4 mL`

ANSWER

`Draw up 0.4 mL`

Find the concentration per mL first

Many paramedics simplify every vial to **"per 1 mL"** first, then divide the order by that number. It is fast and harder to get wrong.

CONCENTRATION **FORMULA 4.2**

$$\text{Concentration (per mL)} = \frac{\text{total drug}}{\text{total volume}}$$

$$\text{Volume (mL)} = \frac{\text{ordered dose}}{\text{concentration per mL}}$$

DRUG AS SUPPLIED	PER ML	ORDER	VOLUME
Epinephrine 1 mg / 10 mL	0.1 mg/mL	1 mg IV (arrest)	10 mL
Epinephrine 1 mg / 1 mL	1 mg/mL	0.3 mg IM	0.3 mL
Amiodarone 150 mg / 3 mL	50 mg/mL	300 mg IV	6 mL
Adenosine 6 mg / 2 mL	3 mg/mL	12 mg rapid IV	4 mL
Atropine 1 mg / 10 mL	0.1 mg/mL	1 mg IV	10 mL
Ondansetron 4 mg / 2 mL	2 mg/mL	4 mg IV	2 mL
Diphenhydramine 50 mg / 1 mL	50 mg/mL	25 mg IV	0.5 mL
Midazolam 10 mg / 2 mL	5 mg/mL	2.5 mg IV	0.5 mL

WATCH THE TOTAL VOLUME ON THE LABEL

"Fentanyl 100 mcg/2 mL" is 50 mcg per mL, not 100. Misreading the total amount as the per-mL concentration is one of the most common drug-math errors. Midazolam comes as both 1 mg/mL and 5 mg/mL; check which one you are holding.

Practice 4: Dose and volume

Answers: Key 4

4.1 Order: diphenhydramine 25 mg IV. Vial: 50 mg/mL. _____ mL

4.2 Order: adenosine 12 mg rapid IV push. Vial: 6 mg/2 mL. _____ mL

4.3 Order: amiodarone 150 mg IV. Vial: 150 mg/3 mL. _____ mL

4.4 Order: fentanyl 50 mcg IN. Vial: 100 mcg/2 mL. _____ mL

4.5 Order: midazolam 2 mg IV. Vial: 5 mg/mL. _____ mL

4.6 Order: naloxone 0.5 mg IV. Vial: 0.4 mg/mL. _____ mL

4.7 Order: ketorolac 15 mg IV. Vial: 30 mg/mL. _____ mL

4.8 Order: epinephrine 0.5 mg IM. Vial: 1 mg/mL. _____ mL

4.9 Order: ondansetron 4 mg IV. Vial: 2 mg/mL. _____ mL

4.10 Order: hydromorphone 500 mcg IV. Vial: 2 mg/mL. _____ mL

CHAPTER FIVE

05 Weight-based calculations

Children, and many adult drugs, are dosed per kilogram. The order reads like mg/kg or mcg/kg. This adds one step in front of the dose-volume formula: find the patient's dose from their weight.

WEIGHT-BASED DOSE — **FORMULA 5.1**

$$\text{Step 1} \cdot \text{kg} = \frac{\text{lb}}{2.2}$$

$$\text{Step 2} \cdot \text{Dose} = (\text{dose per kg}) \times \text{kg}$$

$$\text{Step 3} \cdot \text{Volume} = \frac{\text{Dose}}{\text{concentration per mL}}$$

$$\text{Step 4} \cdot \text{Compare with the maximum single dose}$$

WORKED EXAMPLE 5.1

Pediatric cardiac arrest: epinephrine 0.01 mg/kg IV/IO. Child weighs 44 lb. Vial: 1 mg/10 mL.

WEIGHT

`44 lb ÷ 2.2 = 20 kg`

DOSE

`0.01 mg/kg × 20 kg = 0.2 mg`

CONC.

`1 mg ÷ 10 mL = 0.1 mg/mL`

VOLUME

`0.2 mg ÷ 0.1 mg/mL = 2 mL`

MAX

Single dose max 1 mg. 0.2 mg is under it.

ANSWER

`0.2 mg = 2 mL` Shortcut: with 0.1 mg/mL epinephrine, the arrest dose is 0.1 mL/kg.

WORKED EXAMPLE 5.2 **Fentanyl 1 mcg/kg IV. 66 lb child. Vial 50 mcg/mL.**

WEIGHT

`66 ÷ 2.2 = 30 kg`

DOSE

`1 × 30 = 30 mcg`

VOLUME

`30 ÷ 50 = 0.6 mL`

ANSWER

`0.6 mL`

WORKED EXAMPLE 5.3

Hypoglycemia: D10 0.5 g/kg IV. 10 kg toddler.

DOSE

`0.5 g × 10 = 5 g`

CONC.

`D10 = 10 g/100 mL = 0.1 g/mL`

VOLUME

`5 ÷ 0.1 = 50 mL`

ANSWER

`50 mL of D10` (= 5 mL/kg)

WORKED EXAMPLE 5.4

When the calculation passes the maximum: naloxone 0.1 mg/kg IV, max 2 mg. Child weighs 25 kg. Vial 1 mg/mL.

DOSE

`0.1 × 25 = 2.5 mg`

MAX

2.5 mg is more than the 2 mg maximum. **Give the maximum.**

VOLUME

`2 mg ÷ 1 mg/mL = 2 mL`

ANSWER

`2 mg = 2 mL`

FIELD TIP: LENGTH-BASED TAPES

For children, a length-based resuscitation tape estimates weight from height and gives pre-calculated doses by color zone. Use it, but understand the math behind it so you can catch an error or work without one.

COMMON PEDIATRIC WEIGHT-BASED DOSES	PER KG	USUAL MAX SINGLE DOSE
Epinephrine (arrest), 0.1 mg/mL	0.01 mg/kg (0.1 mL/kg)	1 mg
Epinephrine (anaphylaxis IM), 1 mg/mL	0.01 mg/kg	0.3 mg (child) · 0.5 r
Amiodarone (VF/pVT)	5 mg/kg	300 mg
Adenosine (first dose)	0.1 mg/kg	6 mg
Atropine (bradycardia)	0.02 mg/kg	0.5 mg
Dextrose	0.5–1 g/kg	25 g
Fluid bolus, isotonic	20 mL/kg	Reassess each
Defibrillation	2 J/kg, then 4 J/kg	10 J/kg or adult dose

Reference values from current pediatric resuscitation guidelines; confirm against your protocol.

Practice 5: Weight-based

Answers: Key 5

5.1 154 lb adult. Fentanyl 1 mcg/kg IV. Vial 100 mcg/2 mL. Dose _____ mcg · Volume _____ mL

5.2 33 lb child in cardiac arrest. Epinephrine 0.01 mg/kg, vial 1 mg/10 mL. Dose _____ mg · Volume _____ mL

5.3 20 kg child in VF. Amiodarone 5 mg/kg, vial 150 mg/3 mL. Dose _____ mg · Volume _____ mL

5.4 12 kg child, glucose 38 mg/dL. D10 0.5 g/kg. Dose _____ g · Volume _____ mL

5.5 8 kg infant, symptomatic bradycardia after epinephrine. Atropine 0.02 mg/kg, vial 0.1 mg/mL. Dose _____ mg · Volume _____ mL

5.6 18 kg child, 20 mL/kg NS bolus. _____ mL

5.7 220 lb adult. Ketamine 0.3 mg/kg IV for pain, vial 50 mg/mL. Dose _____ mg · Volume _____ mL

5.8 14 kg child with SVT. Adenosine 0.1 mg/kg, vial 6 mg/2 mL. Dose _____ mg · Volume _____ mL

CHAPTER SIX

06 Drip rates I: volume over time

Volume-based drip problems give you a volume of fluid and a time to run it over. Without a pump, you control the rate by counting drops per minute in the drip chamber. With a pump, you enter mL/hr.

GRAVITY DRIP RATE — **FORMULA 6.1**

$$\text{gtt/min} = \frac{\text{volume (mL)} \times \text{drop factor (gtt/mL)}}{\text{time (min)}}$$

$$\text{mL/hr} = \frac{\text{volume (mL)}}{\text{time (hr)}}$$

drop factor from the tubing package: 10, 15, 20 or 60 **time** in minutes for gtt/min

WORKED EXAMPLE 6.1 Infuse 250 mL NS over 30 minutes with a 10 gtt/mL set

SET UP

(250 mL × 10 gtt/mL) ÷ 30 min

SOLVE

2,500 ÷ 30 = 83.3

ANSWER

83 gtt/min ≈ 21 drops every 15 seconds

WORKED EXAMPLE 6.2

Infuse 1,000 mL LR over 8 hours with a 15 gtt/mL set

MINUTES

8 hr × 60 = 480 min

SOLVE

(1,000 × 15) ÷ 480 = 31.25

PUMP RATE

1,000 ÷ 8 = 125 mL/hr

ANSWER

31 gtt/min or **125 mL/hr** on a pump

Shortcuts that come from the formula

DROP FACTOR	DIVIDE ML/HR BY	SO 120 ML/HR IS
60 gtt/mL	1 (gtt/min = mL/hr)	120 gtt/min
20 gtt/mL	3	40 gtt/min
15 gtt/mL	4	30 gtt/min
10 gtt/mL	6	20 gtt/min

The divisor is 60 ÷ drop factor. It works because gtt/min = (mL/hr × drop factor) ÷ 60.

INFUSION TIME — **FORMULA 6.2**

$$\text{Time (hr)} = \frac{\text{volume (mL)}}{\text{rate (mL/hr)}}$$

Example: 500 mL at 125 mL/hr lasts 4 hours. Useful for planning long transfers and knowing when a bag will run dry.

FIELD TIP: COUNTING DROPS

Count drops for 15 seconds and multiply by 4. To set 83 gtt/min, aim for about 21 drops in 15 seconds. Recheck after moving the patient: bag height, arm position and kinks all change a gravity rate.

IS THE RATE COUNTABLE?

Above about 100–150 gtt/min the drops blur into a stream and can't be counted. If your answer is 300 gtt/min, switch to a set with a smaller drop factor, use a pump, or run it wide open with a pressure bag as protocol allows.

Practice 6: Volume drip rates

Answers: Key 6

6.1 1,000 mL over 4 hours, 15 gtt/mL set. _____ gtt/min · _____ mL/hr

6.2 500 mL over 1 hour, 10 gtt/mL set. _____ gtt/min

6.3 100 mL over 20 minutes, 10 gtt/mL set. _____ gtt/min

6.4 KVO at 30 mL/hr with a 60 gtt/mL set. _____ gtt/min

6.5 250 mL bolus over 15 minutes with a 10 gtt/mL set. _____ gtt/min

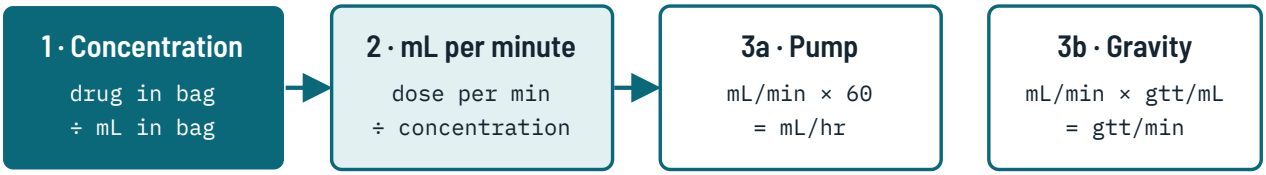
6.6 How long will 1,000 mL last at 125 mL/hr? _____ hr

6.7 A 20 gtt/mL set is running at 25 gtt/min. What is the rate in mL/hr? _____ mL/hr

CHAPTER SEVEN

07 Drip rates II: concentration infusions

A medication infusion is ordered as a dose over time, such as `mcg/min`, `mg/min` or `mcg/kg/min`. To run it you need to know how much drug is in each mL of the bag (the **concentration**), then turn the dose per minute into mL per minute.



Check that the drug units match at step 2: mg with mg, mcg with mcg. Convert the concentration if they don't.

FIGURE 7.1 Every concentration infusion follows the same path.

CONCENTRATION INFUSION **FORMULA 7.1**

$$\text{mL/hr} = \frac{\text{dose per min} \times 60}{\text{concentration (per mL)}}$$

$$\text{gtt/min} = \frac{\text{dose per min} \times \text{drop factor}}{\text{concentration (per mL)}}$$

$$\text{Weight-based: dose per min} = (\text{dose per kg per min}) \times \text{kg}$$

WORKED EXAMPLE 7.1

Norepinephrine 8 mcg/min. Bag: 4 mg in 250 mL. 60 gtt/mL set or pump.

CONVERT

4 mg = 4,000 mcg (order is in mcg)

CONC.

4,000 mcg ÷ 250 mL = 16 mcg/mL

ML/MIN

8 mcg/min ÷ 16 mcg/mL = 0.5 mL/min

PUMP

0.5 × 60 = 30 mL/hr

GRAVITY

0.5 × 60 gtt/mL = 30 gtt/min

ANSWER

30 mL/hr = **30 gtt/min** on a microdrip

WORKED EXAMPLE 7.2

Dopamine 5 mcg/kg/min for an 80 kg patient. Premix: 400 mg in 250 mL.

DOSE/MIN

`5 mcg/kg/min × 80 kg = 400 mcg/min`

CONC.

`400,000 mcg ÷ 250 mL = 1,600 mcg/mL`

ML/MIN

`400 ÷ 1,600 = 0.25 mL/min`

PUMP

`0.25 × 60 = 15 mL/hr`

ANSWER

`15 mL/hr` = **`15 gtt/min`** on a microdrip

WORKED EXAMPLE 7.3

Timed infusion: amiodarone 150 mg in 100 mL D5W over 10 minutes, 10 gtt/mL set

IDEA

The whole bag runs in 10 minutes, so this is really a volume-over-time problem.

GRAVITY

`(100 mL × 10) ÷ 10 min = 100 gtt/min`

PUMP

`100 mL ÷ (10/60 hr) = 600 mL/hr, VTBI 100 mL`

ANSWER

`100 gtt/min` or **`600 mL/hr`**

Working backward: what dose is running?

On a transfer you often inherit a running drip. Reverse the formula to confirm the dose matches the order.

DOSE FROM RATE **FORMULA 7.2**

$$\text{dose per min} = \frac{\text{mL/hr} \times \text{concentration}}{60}$$

Example: norepinephrine 16 mcg/mL running at 15 mL/hr → 15 × 16 ÷ 60 = 4 mcg/min.

Standard concentrations at a glance

DRUG (MIX)	CONCENTRATION	RATE PER DOSE STEP
Norepinephrine 4 mg / 250 mL	16 mcg/mL	1 mcg/min = 3.75 mL/hr
Epinephrine 1 mg / 250 mL	4 mcg/mL	1 mcg/min = 15 mL/hr
Dopamine 400 mg / 250 mL	1,600 mcg/mL	mL/hr = mcg/kg/min × kg ÷ 26.7
Lidocaine 2 g / 500 mL	4 mg/mL	1 mg/min = 15 mL/hr
Amiodarone 450 mg / 250 mL	1.8 mg/mL	1 mg/min = 33.3 mL/hr
Magnesium 2 g / 50 mL	40 mg/mL	Over 15 min = 200 mL/hr

Many services use pharmacy premixed bags; concentrations differ between systems. Always read the bag label.

FIELD TIP: THE CLOCK METHOD

For any drug mixed at 4 units per mL (lidocaine 4 mg/mL, epinephrine 4 mcg/mL) on a 60 gtt/mL set, 1 unit/min = 15 gtt/min, 2 = 30, 3 = 45, 4 = 60. Picture the quarter-hours on a clock face.

VASOPRESSORS ARE HIGH-ALERT

Run vasopressors on a pump through a dedicated line whenever possible, label the line, and never flush it: a flush pushes a bolus of drug. Watch the site closely for extravasation.

Practice 7: Concentration infusions

Answers: Key 7

7.1 Norepinephrine 12 mcg/min. Bag 4 mg/250 mL. 60 gtt/mL set. Concentration _____ · _____ gtt/min

7.2 Dopamine 10 mcg/kg/min, 70 kg patient. Bag 400 mg/250 mL. _____ mL/hr

7.3 Epinephrine 5 mcg/min. Bag 1 mg/250 mL. _____ mL/hr

7.4 Magnesium sulfate 2 g in 50 mL over 15 minutes, 10 gtt/mL set. _____ gtt/min · pump _____ mL/hr

7.5 Lidocaine 3 mg/min. Bag 2 g/500 mL. 60 gtt/mL set. _____ gtt/min

7.6 You receive a patient with norepinephrine 4 mg/250 mL running at 15 mL/hr. What dose is the patient getting? _____ mcg/min

7.7 Amiodarone maintenance 1 mg/min. Bag 450 mg/250 mL. _____ mL/hr

7.8 Dopamine 400 mg/250 mL is running at 30 mL/hr on a 75 kg patient. What is the dose in mcg/kg/min? _____

CHAPTER EIGHT

08 Percent and ratio concentrations

Some drugs are labeled by percentage (lidocaine 2%, D50, magnesium 50%) or by an older ratio (epinephrine 1:1,000). Before you can calculate a volume, convert the label to **mg per mL** or **g per mL**.

PERCENT (WEIGHT/VOLUME) — **FORMULA 8.1**

$$X\% = X \text{ g per } 100 \text{ mL}$$

$$\text{mg/mL} = \% \times 10$$

1% = 1 g/100 mL = 1,000 mg/100 mL = 10 mg/mL

RATIO — **FORMULA 8.2**

$$1{:}X = 1 \text{ g in } X \text{ mL}$$

$$1{:}1{,}000 = 1 \text{ mg/mL}$$

$$1{:}10{,}000 = 0.1 \text{ mg/mL}$$

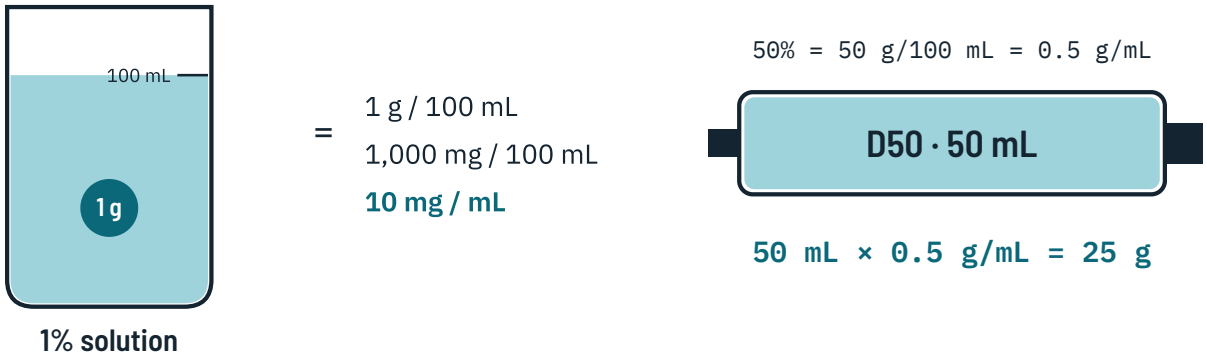


FIGURE 8.1 Percent means grams per 100 mL. Multiply by 10 to get mg/mL.

LABEL	MEANS	PER ML	EXAMPLE DOSE → VOLUME
Lidocaine 2%	2 g/100 mL	20 mg/mL	100 mg → 5 mL
Dextrose 50% (D50)	50 g/100 mL	0.5 g/mL	25 g → 50 mL
Dextrose 25% (D25)	25 g/100 mL	0.25 g/mL	5 g → 20 mL
Dextrose 10% (D10)	10 g/100 mL	0.1 g/mL	25 g → 250 mL
Magnesium sulfate 50%	50 g/100 mL	500 mg/mL	2 g → 4 mL
Calcium chloride 10%	10 g/100 mL	100 mg/mL	1 g → 10 mL
Calcium gluconate 10%	10 g/100 mL	100 mg/mL	3 g → 30 mL
Sodium bicarbonate 8.4%	8.4 g/100 mL	84 mg = 1 mEq/mL	50 mEq → 50 mL
Sodium chloride 0.9%	0.9 g/100 mL	9 mg/mL	1 L contains 9 g NaCl
Epinephrine 1 mg/mL (old 1:1,000)	1 g/1,000 mL	1 mg/mL	0.3 mg IM → 0.3 mL
Epinephrine 0.1 mg/mL (old 1:10,000)	1 g/10,000 mL	0.1 mg/mL	1 mg IV → 10 mL

CALCIUM CHLORIDE VS. CALCIUM GLUCONATE

Both are 10%, but calcium chloride delivers about three times more elemental calcium per mL. They are not interchangeable mL for mL. Confirm which one the protocol names.

WHY RATIO LABELS CHANGED

Since 2016 US labels state epinephrine as mg/mL (1 mg/mL, 0.1 mg/mL) instead of 1:1,000 and 1:10,000, because ratio labels caused 10-fold errors. You will still hear the old names on calls and see them in older references.

Dilutions: $C_1V_1 = C_2V_2$

When you need a weaker solution than you have, use the dilution formula. The amount of drug stays the same; only the volume grows.

DILUTION **FORMULA 8.3**

$$C_1 \times V_1 = C_2 \times V_2$$

C_1 V_1 starting concentration and volume **C_2 V_2** final concentration and total volume

WORKED EXAMPLE 8.1

Make D10 from a 50 mL D50 syringe

SET UP

$50\% \times 50\ mL = 10\% \times V_2$

SOLVE

$V_2 = 2{,}500 \div 10 = 250\ mL$

MIX

Add 50 mL D50 to 200 mL NS or sterile water.

ANSWER

250 mL of D10 (25 g)

Rule of thumb: D50 to D10 is 1 part D50 + 4 parts diluent. D50 to D25 is 1 + 1.

WORKED EXAMPLE 8.2 Push-dose epinephrine 10 mcg/mL

START

`0.1 mg/mL epi = 100 mcg/mL`

SET UP

$100 \text{ mcg/mL} \times V_1 = 10 \text{ mcg/mL} \times 10 \text{ mL}$

SOLVE

$V_1 = 1 \text{ mL}$

MIX

Waste 1 mL from a 10 mL NS flush, add 1 mL of 0.1 mg/mL epinephrine. Label it.

ANSWER

10 mL at 10 mcg/mL

Practice 8: Percent and ratio

Answers: Key 8

8.1 Lidocaine 2%: how many mg/mL? _____ How many mL for 1.5 mg/kg in a 60 kg patient? _____ mL

8.2 Magnesium sulfate 50%, order 2 g IV. _____ mL

8.3 How many grams of dextrose are in 250 mL of D10? _____ g

8.4 Calcium chloride 10%, order 500 mg IV. _____ mL

8.5 Sodium bicarbonate 8.4%, order 1 mEq/kg for a 70 kg patient. _____ mL

8.6 You have 25 mL of D50. How much diluent makes D10, and what is the total volume? _____ mL diluent · _____ mL total

8.7 Epinephrine 1:1,000, order 0.5 mg IM. _____ mL

8.8 How many grams of NaCl are in 500 mL of 0.9% NS? _____ g

8.9 Push-dose epinephrine 10 mcg/mL, order 20 mcg. _____ mL

CHAPTER NINE

09 IV infusion pumps

A volumetric infusion pump pushes fluid at a set **rate in mL/hr**, regardless of bag height, back-pressure or patient position. That makes it the standard for medication infusions, children, and any transfer where the rate must be exact. Pumps differ by brand, but the controls and concepts below are common to all of them. Learn your service's model from its manufacturer training.

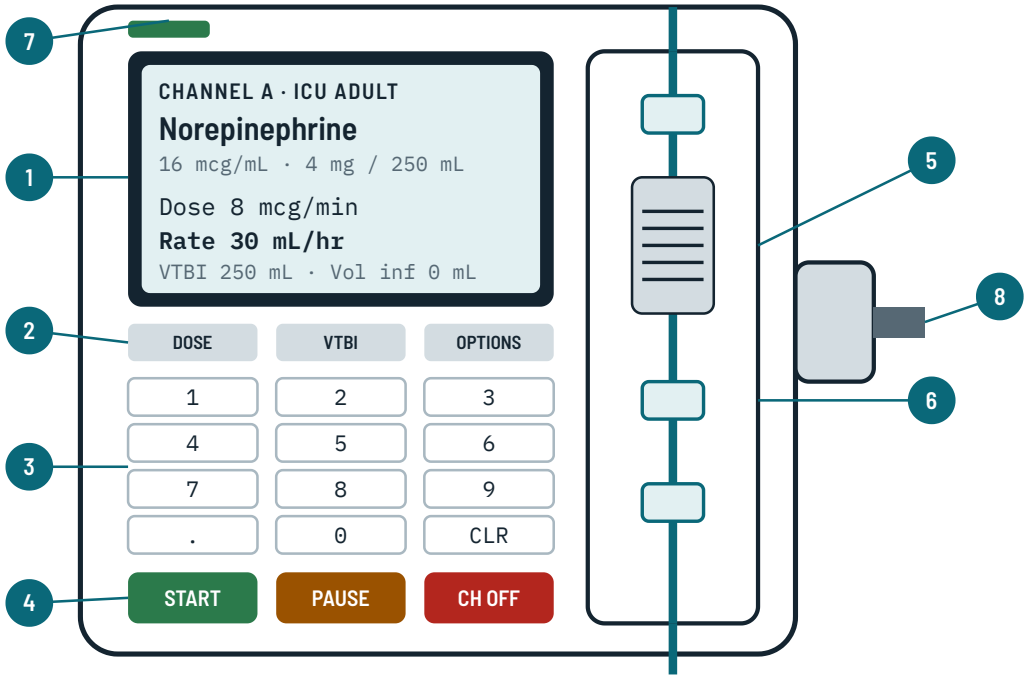


FIGURE 9.1 A generic large-volume infusion pump. Layout varies by manufacturer, but every pump has these parts.

#	PART	WHAT IT DOES
1	Display	Shows drug, concentration, dose, rate (mL/hr), VTBI, volume infused, alarms and battery.
2	Soft keys	Context buttons whose labels change on screen: dose, VTBI, options, secondary.
3	Numeric keypad	Enters rate, dose, weight and volume. Watch the decimal key.
4	Start · Pause · Channel off	Start runs the program. Pause holds it. Channel off stops it and clears that channel.
5	Door and set	The pump segment of the tubing loads here. The pumping mechanism (peristaltic fingers or a cassette) moves fluid. Anti-free-flow clamps close when the door opens.
6	Sensors	Upstream and downstream pressure sensors detect occlusions; the air-in-line detector catches bubbles.
7	Power / battery	Shows AC or battery operation and charge left. Plan for transport time.
8	Pole clamp	Mounts the pump to an IV pole or stretcher mount.

Key terms

Rate	Volume per hour, in `mL/hr`. The pump always runs on this, even in dose mode.
VTBI	Volume to be infused. The pump stops (or drops to KVO) when it reaches this. Set it to the volume in the bag or the volume ordered, whichever is less.
Volume infused	Running total. Record it at handoff and clear it at the start of your transport.
Rate mode vs. dose mode	Rate mode: you enter mL/hr. Dose mode: you enter drug, concentration, weight and dose, and the pump calculates mL/hr. Verify the calculation yourself.
Drug library (DERS)	Dose error reduction software: preloaded drugs, standard concentrations and dose limits by care area. Always choose the drug from the library instead of using basic rate mode.
Soft limit	An unusual dose. The pump warns you; you can override after re-checking.
Hard limit	An unsafe dose. The pump will not accept it. Re-check the order and the math.
KVO	Keep vein open: a low rate (often 1–5 mL/hr) the pump falls back to when VTBI is complete.
Primary vs. secondary	Primary is the main infusion. Secondary (piggyback) runs a short infusion through the same channel, then returns to primary.
Bolus / loading dose	A set amount delivered over a set time from the same bag, on pumps that support it.

Programming a medication infusion, step by step

1. **Verify the order and the rights.** Drug, concentration, dose, patient weight, route, and any titration parameters.
2. **Calculate the expected rate by hand.** You need your own number to check the pump against (Chapter 7).
3. **Spike and prime the set.** Clear all air, label the tubing with the drug name, and close the roller clamp.

4. **Power on and choose the care area or profile.** This sets which drug library and limits apply.
5. **Load the set into the channel and close the door.** Orient the set in the direction of flow (bag above, patient below).
6. **Select the drug from the library.** Confirm the concentration on screen matches the bag label exactly.
7. **Enter weight (if weight-based), dose and VTBI.** Read the rate the pump calculated.
8. **Compare the pump's rate to your hand calculation.** If they differ, stop and find out why before you start.
9. **Independent double-check with your partner.** They check drug, concentration, dose, rate, line and patient on their own.
10. **Open the roller clamp, connect to the patient, press Start.** Trace the line from bag to patient to confirm the right channel and port.
11. **Monitor.** Watch the drip chamber, vital signs and IV site. Document start time, dose and rate.

Running a secondary (piggyback) infusion

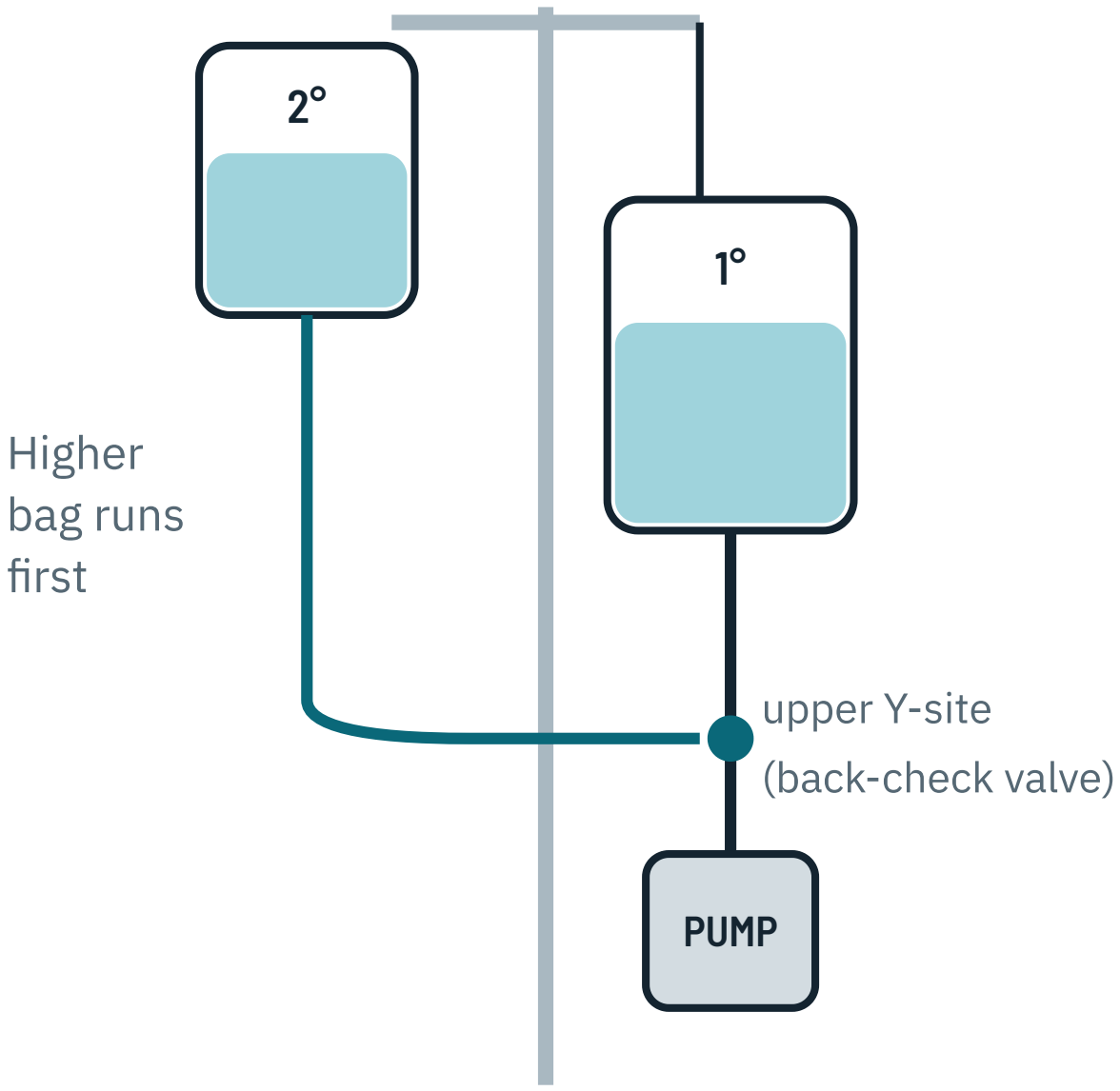


FIGURE 9.2 The secondary bag must hang higher than the primary.

1. Hang the secondary bag higher than the primary; use the extension hook to lower the primary.
2. Connect the secondary line to the **upper** Y-site, above the pump.
3. Select **Secondary** on the pump. Enter the drug, rate and VTBI for the secondary.
4. Open the secondary clamp. Watch the secondary drip chamber to confirm it is the one dripping.
5. When VTBI is complete the pump returns to the primary rate. Confirm the primary resumed.

COMMON MISS

If the secondary clamp is closed, or the bags are at the same height, the pump runs *primary* fluid at the secondary rate. Always look at which chamber is dripping.

Alarm troubleshooting

ALARM	LIKELY CAUSE	WHAT TO DO
Downstream occlusion (patient side)	Closed clamp below the pump, kinked line, patient's arm bent, infiltrated or clotted IV	Trace the line to the patient. Open clamps, straighten the arm, check and flush the site (not a vasopressor line).
Upstream occlusion (bag side)	Roller clamp closed above the pump, empty bag, vent closed on a glass bottle, kink	Open the clamp, check the bag and the spike, straighten the tubing.
Air in line	Bubbles from poor priming, empty bag, loose connection	Pause, clamp, remove air per the manufacturer's method, re-prime. Never just silence and restart.
Infusion complete / VTBI reached	Programmed volume delivered; pump is at KVO	Hang a new bag or reprogram if the order continues.
Door open / set not loaded	Set not seated, door not latched	Reload the set in the direction of flow and close the door.
Low battery	Running on battery too long	Plug into AC or the ambulance inverter. Check charge before departure.
Channel paused too long (callback)	Program left on pause	Resume or cancel. Don't leave a drip paused unnoticed.
Dose limit (soft or hard)	Entered dose outside library limits	Re-check the order, units, weight and concentration. Soft limits allow an override only after verification.

Pumps on interfacility transports

- **Verify on arrival.** Match each channel against the transfer orders: drug, concentration, dose, rate, line and port. Record volume infused, then clear it.
- **Count the bags.** Will each bag and the battery last the trip plus delays? Use Formula 6.2.
- **Move one thing at a time.** When switching from the hospital's pump to yours, clamp the line before opening any door so nothing free-flows, reprogram, re-verify, then unclamp.
- **Stay in scope.** Only run and titrate medications your protocols and the transfer orders allow.
- **Hand off by reading back.** At the receiving facility, read each pump's settings aloud with the receiving clinician.

Practice 9: Infusion pumps

Answers: Key 9

9.1 Define VTBI. If a 250 mL bag is ordered to run 100 mL, what do you set as VTBI? _____ mL

9.2 Explain the difference between a soft limit and a hard limit in a drug library.

9.3 The pump alarms "downstream occlusion" right after the patient bends their elbow. What do you check first?

9.4 You program magnesium as a secondary, but the primary NS chamber is the one dripping. Name two likely causes.

9.5 Norepinephrine 4 mg/250 mL at 6 mcg/min is ordered. The pump in dose mode shows 22.5 mL/hr. Is that correct? Show your check.

9.6 A 500 mL bag is running at 75 mL/hr with 320 mL left. Your transport is 3.5 hours. Will it last?

REFERENCE

10 Formula reference card

Every formula in this workbook on one page. Copy it into your field guide.

METRIC

kg ↔ g ↔ mg ↔ mcg: × or ÷ 1,000

1 L = 1,000 mL

WEIGHT

kg = lb ÷ 2.2

lb = kg × 2.2

DOSE TO VOLUME

mL = (D ÷ H) × Q

mL = dose ÷ (amount per mL)

WEIGHT-BASED DOSE

dose = dose/kg × kg

then check the max dose

GRAVITY DRIP

gtt/min = mL × gtt/mL ÷ min

gtt/min = mL/hr ÷ (60 ÷ gtt/mL)

PUMP RATE (VOLUME)

mL/hr = mL ÷ hr

time (hr) = mL ÷ mL/hr

CONCENTRATION

conc = drug in bag ÷ mL in bag

MEDICATION INFUSION

mL/hr = dose/min × 60 ÷ conc

gtt/min = dose/min × gtt/mL ÷ conc

DOSE FROM RATE

dose/min = mL/hr × conc ÷ 60

WEIGHT-BASED INFUSION

mL/hr = mcg/kg/min × kg × 60 ÷ conc

PERCENT

X% = X g / 100 mL

mg/mL = % × 10

RATIO AND DILUTION

1:1,000 = 1 mg/mL · 1:10,000 = 0.1 mg/mL

$C_1V_1 = C_2V_2$

FLUIDS

Peds bolus 20 mL/kg

Burns 2–4 mL × kg × %TBSA / 24 h

TEMPERATURE

°C = (°F − 32) ÷ 1.8

°F = °C × 1.8 + 32

ANSWER KEY

11 Worked solutions

Try every problem before you check. If your answer differs, find the step where it went wrong. That step is what to practice.

PRACTICE 2 · CONVERSIONS

2.1 0.4 × 1,000 = **400 mcg**

2.2 2,500 ÷ 1,000 = **2.5 mg**

2.3 198 ÷ 2.2 = **90 kg**

2.4 1.2 × 1,000 = **1,200 mL**

2.5 3.5 × 2.2 = **7.7 lb**

2.6 0.025 × 1,000 = **25 mg**

2.7 39.5 × 1.8 = 71.1; + 32 = **103.1 °F**

2.8 90 ÷ 60 = **1.5 mL/min**

PRACTICE 3 · IV FLUIDS

3.1 **0.9% NS**. LR contains calcium, which binds the citrate anticoagulant in stored blood and can cause clotting in the line.

3.2 The dextrose is taken up and metabolized, leaving free water that moves into cells.

3.3 **Isotonic · hypotonic · hypertonic**

3.4 20 × 25 = **500 mL**

3.5 Hypotonic fluids (D5W, 0.45% NaCl): free water shifts into brain cells and worsens cerebral edema. Many protocols also prefer NS over LR.

3.6 4 × 60 × 40 = 9,600 mL; half = 4,800 mL over 8 h = **600 mL/hr**

3.7 About 1/4: **≈ 250 mL**

PRACTICE 4 · DOSE AND VOLUME

4.1 25 ÷ 50 × 1 = **0.5 mL**

4.2 12 ÷ 6 × 2 = **4 mL**

4.3 150 ÷ 150 × 3 = **3 mL**

4.4 50 ÷ 100 × 2 = **1 mL**

4.5 2 ÷ 5 × 1 = **0.4 mL**

4.6 0.5 ÷ 0.4 × 1 = **1.25 mL** (1.3 mL on a 3 mL syringe)

4.7 15 ÷ 30 × 1 = **0.5 mL**

4.8 0.5 ÷ 1 × 1 = **0.5 mL**

4.9 4 ÷ 2 = **2 mL**

4.10 500 mcg = 0.5 mg; 0.5 ÷ 2 = **0.25 mL**

PRACTICE 5 · WEIGHT-BASED

5.1 70 kg → **70 mcg** ; 50 mcg/mL → **1.4 mL**

5.2 15 kg × 0.01 = **0.15 mg** ; ÷ 0.1 = **1.5 mL**

5.3 5 × 20 = **100 mg** ; 50 mg/mL → **2 mL**

5.4 0.5 × 12 = **6 g** ; ÷ 0.1 g/mL = **60 mL**

5.5 0.02 × 8 = **0.16 mg** ; ÷ 0.1 = **1.6 mL**

5.6 20 × 18 = **360 mL**

5.7 220 ÷ 2.2 = 100 kg × 0.3 = **30 mg** ; ÷ 50 = **0.6 mL**

5.8 0.1 × 14 = **1.4 mg** (under 6 mg max); 3 mg/mL → **0.47 mL**

PRACTICE 6 · VOLUME DRIP RATES

6.1 (1,000 × 15) ÷ 240 = 62.5 → **63 gtt/min** ; 1,000 ÷ 4 = **250 mL/hr**

6.2 (500 × 10) ÷ 60 = 83.3 → **83 gtt/min**

6.3 (100 × 10) ÷ 20 = **50 gtt/min**

6.4 Microdrip: gtt/min = mL/hr → **30 gtt/min**

6.5 (250 × 10) ÷ 15 = 166.7 → **167 gtt/min** . Too fast to count: use a pump or run wide open and watch the bag.

6.6 1,000 ÷ 125 = **8 hr**

6.7 25 gtt/min ÷ 20 gtt/mL = 1.25 mL/min × 60 = **75 mL/hr**

PRACTICE 7 · CONCENTRATION INFUSIONS

7.1 **16 mcg/mL** ; 12 ÷ 16 = 0.75 mL/min × 60 = **45 gtt/min**

7.2 10 × 70 = 700 mcg/min ÷ 1,600 = 0.4375 mL/min × 60 = **26.3 mL/hr**

7.3 4 mcg/mL; 5 ÷ 4 = 1.25 mL/min × 60 = **75 mL/hr**

7.4 (50 × 10) ÷ 15 = 33.3 → **33 gtt/min** ; 50 ÷ 0.25 hr = **200 mL/hr**

7.5 4 mg/mL; 3 ÷ 4 = 0.75 mL/min × 60 = **45 gtt/min**

7.6 15 × 16 ÷ 60 = **4 mcg/min**

7.7 1.8 mg/mL; 1 ÷ 1.8 = 0.556 mL/min × 60 = **33.3 mL/hr**

7.8 30 × 1,600 ÷ 60 = 800 mcg/min ÷ 75 kg = **10.7 mcg/kg/min**

PRACTICE 8 · PERCENT AND RATIO

8.1 **20 mg/mL**; 1.5 × 60 = 90 mg ÷ 20 = **4.5 mL**

8.2 500 mg/mL; 2,000 ÷ 500 = **4 mL**

8.3 250 × 0.1 = **25 g**

8.4 100 mg/mL; 500 ÷ 100 = **5 mL**

8.5 70 mEq at 1 mEq/mL = **70 mL**

8.6 50% × 25 = 10% × V_2 → V_2 = 125 mL: **100 mL diluent · 125 mL total** (12.5 g)

8.7 1 mg/mL → **0.5 mL**

8.8 0.9 g/100 mL × 5 = **4.5 g**

8.9 20 ÷ 10 = **2 mL**

PRACTICE 9 · INFUSION PUMPS

9.1 The volume the pump will deliver before stopping or going to KVO. **100 mL**

9.2 Soft limit: a warning you may override after verifying. Hard limit: the pump refuses the entry.

9.3 The patient side: straighten the arm, look for a closed clamp or kink, then check the IV site for infiltration.

9.4 Secondary clamp closed; secondary bag not hung higher than the primary (also: connected below the pump).

9.5 6 ÷ 16 = 0.375 mL/min × 60 = **22.5 mL/hr: correct**

9.6 320 ÷ 75 = 4.27 hr. **Yes**, with about 45 minutes of margin. Bring a spare bag anyway.

This workbook is an educational resource for paramedic students. Drug concentrations, doses and fluid volumes are examples that reflect common EMS practice and current resuscitation guidance at the time of writing; they are not orders. Always follow your local protocols, your medical director, the product labeling, and manufacturer instructions for your equipment.